

Comparative Analysis of PSO, HHO, and GA for Parameter Estimation in Lithium-Ion Battery Thevenin Models

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Abstract—Lithium-ion battery modeling is important for energy storage systems such as electric vehicles and renewable energy sources. This paper presents a comparison of three metaheuristic optimization algorithms, namely Particle Swarm Optimization (PSO), Harris Hawks Optimization (HHO), and Genetic Algorithm (GA), to estimate the parameters of R_0 (internal resistance), R_1 (polarization resistance), and C_1 (capacitance) for a first-order Thevenin equivalent circuit model. The authors generated a synthetic dataset using known parameter values and ran each algorithm under the same constraints and number of iterations. Three accuracy metrics root mean square error (RMSE), mean absolute error (MAE), and maximum absolute error (MaxAE) were used to evaluate the estimation performance. The results indicate that all three algorithms achieved accurate predictions, with PSO delivering the best performance by perfectly estimating all parameters: $R_0 = 0.015, \Omega$, $R_1 = 0.025, \Omega$, and $C_1 = 1500, F$, resulting in RMSE = 0.00000 V, MAE = 0.00000 V, and Max Error = 0.00000 V. HHO also provided near-optimal results with very low error values (e.g., RMSE = 0.00003 V), while GA produced a slightly less accurate fit with a noticeable deviation in C_1 and higher errors (e.g., RMSE = 0.00078 V). According to the findings, metaheuristic algorithms can be effectively applied for modeling nonlinear battery systems and demonstrate competitive performance.

Index Terms—Lithium-ion battery modeling, Thevenin equivalent circuit, Metaheuristic algorithms, Battery Management Systems (BMS), Nonlinear optimization

I. INTRODUCTION

Battery modeling is a wide and interdisciplinary field that deals with problems of chemists, physicists, and electrical engineers. The electrochemical, thermal, and electrical behavior occurring in a battery cell are interdependent and highly nonlinear [11]. Among different modeling techniques, equivalent circuit models (ECMs) are often used because of their structural simplicity, modeling accuracy, and capacity to capture key cell characteristics such as open circuit voltage (OCV), internal resistance, and transient polarization [12]. These models generally consist of voltage sources and passive components (resistors and capacitors) connected in RC pairs, while high-frequency effects are captured with the inclusion

of inductance elements [13]. Energy consumption calculations often apply simplified models; one example is the Rint model. According to [14], battery models with a single internal resistance and voltage source that depends on state of charge (SOC) are common. More complex structures are adopted when modeling dynamic responses, such as the Thevenin model. Identification of parameters in such models is often carried out using the Hybrid Pulse Power Characterization (HPPC) test, which enables the extraction of steady-state and dynamic behaviors [15], [16]. The OCV in Thevenin models is nonlinear with SOC, typically exhibiting three regions: a steep drop at high SOC, a more linear mid-low region, and another steep drop around 0.3 to 0.4 SOC. Careful modeling is necessary, as OCV-SOC data obtained from charge/discharge cycles can be influenced by impedance effects and energy losses, which differ between charging and discharging. To address this, HPPC-based averaging of the no-current voltage before each pulse is proposed for more reliable OCV curve modeling [11]. OCV measurements often show irregularities, complicating their use in simulations. Therefore, approximation methods employing exponential, polynomial, and hybrid functions are commonly utilized [13], [16]. These approximations may use analytical techniques based on manually selected inflection points or optimization algorithms for parameter fitting [17]. Identifying time constants and RC parameters for Thevenin models is another key task in battery modeling. Generally using HPPC or pulse discharge tests, this task involves fitting the voltage response to multi-exponential functions, with each exponential term representing an RC pair in the model. One-time-constant models are simpler and easier to identify, while two-time-constant models offer greater accuracy at a higher complexity cost [18]. Deterministic optimization techniques have been used for parameter estimation, but they are sensitive to initial conditions and prone to local optima. Unlike deterministic methods, metaheuristic algorithms such as Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Harris Hawks Optimization (HHO) can perform global optimization without

relying on gradients. These algorithms explore multiple solutions simultaneously, increasing the likelihood of finding a global optimum [19], [20]. The optimization efficiency of metaheuristic algorithms depends on their cognitive strategies (e.g., global best, local best), swarm topologies (e.g., ring lattice, von Neumann grid), fitness-distance-ratio mechanisms, and tuning parameters (e.g., cognitive coefficient, inertia weight, constriction factor). Choosing suitable configurations is critical for achieving efficient optimization. This research is motivated by the relevance of model-based design (MBD) in power systems, specifically for developing fully electric load-haul-dump (LHD) vehicles. The proposed cell modeling and parameter identification methods will aid in simulating and designing a full traction battery system. MBD-based simulation techniques are increasingly adopted in aerospace, automotive, and robotics industries to improve system reliability and reduce development time [12].

A. Objectives and Contributions

This paper proposes a comparative study of PSO, HHO, and GA for estimating the parameters of a first-order Thevenin model using synthetic discharge data. Each algorithm is tested under identical boundary conditions, and performance is evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Maximum Error metrics in voltage prediction. Convergence trends are also analyzed to assess the efficiency and reliability of each algorithm.

The main contributions of this paper are:

- Implementation and application of three metaheuristic algorithms (PSO, HHO, GA) to battery parameter estimation using a common fitness function and synthetic dataset.
- Performance comparison based on predictive accuracy and convergence behavior.

II. RELATED WORK

Over the past few years, lithium-ion battery modelling using metaheuristic optimization algorithms has gained much attention. Most of the traditional estimation methods, like recursive least squares (RLS) and extended Kalman filters (EKF), rely on linear assumptions and are sensitive to initial conditions. Consequently, nature-inspired algorithms have emerged as effective alternatives for the identification of nonlinear dynamic models of energy storage systems. Castanho et al. (2022) presented a methodology using the Particle Swarm Optimization (PSO) with multiple linear regression to estimate the state of charge (SoC). According to their study, modeling error and terminal voltage deviation can effectively be minimized by PSO. Wang et al. (2021) proposed a hybrid estimation framework based on self-adaptive PSO and Differential Evolution (DE) to estimate parameters for the Thevenin model in various operating ranges. Their approach converges faster and has a smaller RMSE than conventional methods. More recent techniques, such as Harris Hawks Optimization (HHO), are being studied for battery modeling. Karimi et al. (2023) used HHO to optimize a high-power lithium-ion battery model at

various current rates and a large range of temperatures. The results show that HHO is more generalizable and adaptive than PSO, notably in extreme operational cases. Pizarro-Carmona et al. (2021) employed the Genetic Algorithm (GA) to calibrate an equivalent circuit model of a lithium-ion battery pack, proving its effectiveness in minimizing both transient and steady-state voltage deviations. Recent research shows the combination of metaheuristics and adaptive systems for better estimation robustness. Tang et al. (2022) developed a hybrid framework using GA, PSO, and fuzzy logic controllers that could jointly improve the SoC estimation under noisy conditions. Moreover, Jamali and Eftekhari (2021) introduced a variant of PSO featuring adaptive reinitialization, designed to counteract stagnation and premature convergence with promising outcomes. In addition, the stepwise execution of this process enhanced PSO's efficiency on complex nonlinear-surface-requiring optimization problems. Despite various studies showing the benefits of using PSO, HHO, and GA for battery modeling, a unified testing environment for comparison between these studies is still lacking. Many works highlight the performance of their single-method application over varying test cases and therefore their relative performance is low. This study aims to fill the gap in research by comparing PSO, HHO, and GA under the same test conditions using the same synthetic datasets and fitness evaluation metrics, which will help in assessing the applicability of each algorithm to real-world battery management applications.

III. METHODOLOGY

This section presents the methodological framework adopted to identify the key parameters of a first-order Thevenin equivalent circuit model of a lithium-ion battery. The methodology consists of five main phases: (1) generation of synthetic voltage data using predefined true parameters, (2) formulation of an objective function based on voltage error minimization, (3) application of three distinct metaheuristic algorithms—PSO, HHO, and GA, (4) evaluation of the results using statistical error metrics, and (5) performance comparison across the optimization approaches. The complete workflow is illustrated in Figure 1, which shows the sequential steps from synthetic data generation to parameter estimation and final comparative analysis.

A. Synthetic Data Generation

To evaluate the performance of the optimization algorithms under controlled conditions, synthetic terminal voltage data was generated using the Thevenin model. This model captures the internal resistance and dynamic response of the battery and is widely used in state estimation and control applications [12], [13]. The voltage response is simulated under constant current discharge and is given by the following equation:

$$V(t) = V_{OC} - I \cdot R_0 - R_1 \cdot I \cdot \left(1 - e^{-t/(R_1 C_1)}\right) \quad (1)$$

where:

- $V(t)$ is the terminal voltage at time t ,

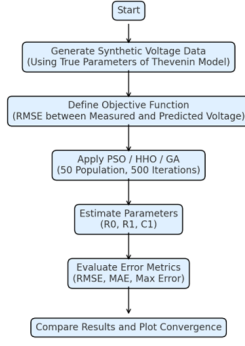


Fig. 1. Methodology workflow

- V_{OC} is the open-circuit voltage (fixed at 3.7 V),
- I is the applied constant current (5 A),
- R_0 is the ohmic resistance,
- R_1 and C_1 represent the dynamic RC network capturing polarization effects.

The ground-truth parameters used for data generation were $R_0 = 0.015, \Omega$, $R_1 = 0.025, \Omega$, and $C_1 = 1500, F$. The voltage data was generated over a 100-second simulation period with a uniform sampling interval, resulting in 500 data points.

B. Objective Function

The optimization objective was to minimize the difference between the synthetic (measured) and predicted terminal voltage across the dataset. This is quantitatively expressed using the Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (V_{\text{measured}}(i) - V_{\text{predicted}}(i))^2} \quad (2)$$

where n is the number of time samples. The RMSE was used as the fitness function in all three optimization algorithms to guide the search toward parameter sets that minimize voltage prediction error.

C. Optimization Algorithms

To solve the parameter estimation problem, three meta-heuristic algorithms were independently employed:

- **Particle Swarm Optimization (PSO):** A population-based algorithm inspired by bird flocking behavior. It adjusts each particle's trajectory based on its own and its neighbors' experience [12]. Configured with 50 particles and 500 iterations.
- **Harris Hawks Optimization (HHO):** Inspired by the cooperative hunting strategy of Harris' hawks, dynamically adjusting strategies based on prey energy [?].
- **Genetic Algorithm (GA):** Evolutionary optimization based on selection, crossover, and mutation, retaining elite solutions per generation [?].

All algorithms were constrained within practical boundaries: $R_0, R_1 \in [0.005, 0.05], \Omega$ and $C_1 \in [500, 5000], F$. These settings balance convergence and computational effort [16].

D. Evaluation Metrics

After optimization, the estimated parameters were used to regenerate the terminal voltage curve and compared to the synthetic voltage. The following metrics were computed:

- **Root Mean Square Error (RMSE)** — as defined in Eq. (2)
- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n} \sum_{i=1}^n |V_{\text{measured}}(i) - V_{\text{predicted}}(i)| \quad (3)$$

- **Maximum Absolute Error:**

$$\text{Max Error} = \max(|V_{\text{measured}} - V_{\text{predicted}}|) \quad (4)$$

These metrics offer interpretability and are widely used in battery model estimation performance evaluations [?], [?].

IV. RESULTS

The results of implementing PSO, HHO, and GA techniques to identify parameter values of the first-order Thevenin equivalent circuit model for lithium-ion batteries are described in this section. Under a constant current discharge situation, the objective is to minimize the RMSE between the predicted terminal voltage and the synthetically generated actual voltage.

A. Parameter Estimation Accuracy

The true parameters utilized to generate the synthetic dataset were $R_0 = 0.015, \Omega$, $R_1 = 0.025, \Omega$, and $C_1 = 1500, F$. All three algorithms were executed with a population size of 50 and a maximum of 500 iterations. The estimated parameters and corresponding error metrics are summarized in Table I.

TABLE I
ESTIMATED PARAMETERS AND ERROR METRICS FOR PSO, HHO, AND GA

Parameter	True Value	PSO	HHO	GA
R_0 (Ω)	0.015	0.01500	0.01502	0.01440
R_1 (Ω)	0.025	0.02500	0.02499	0.02522
C_1 (F)	1500	1500.00	1504.00	1393.00
RMSE (V)	—	0.00000	0.00003	0.00078
MAE (V)	—	0.00000	0.00002	0.00057
Max Error (V)	—	0.00000	0.00012	0.00299

The results revealed that PSO perfectly estimated all parameters, with zero error values. HHO outputs were nearly identical to the true model, with minimal error. GA showed noticeable deviation in estimating C_1 , resulting in higher prediction error.

B. Voltage Prediction Performance

Figure 2 illustrates the voltage prediction performance of the estimated models against the measured voltage curve. Both PSO and HHO closely match the true voltage profile, while the GA-based model shows a slight deviation due to the underestimated capacitance.

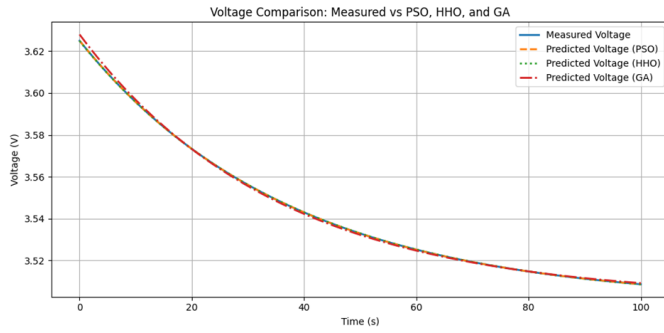


Fig. 2. Comparison between measured and predicted terminal voltages using PSO, HHO, and GA.

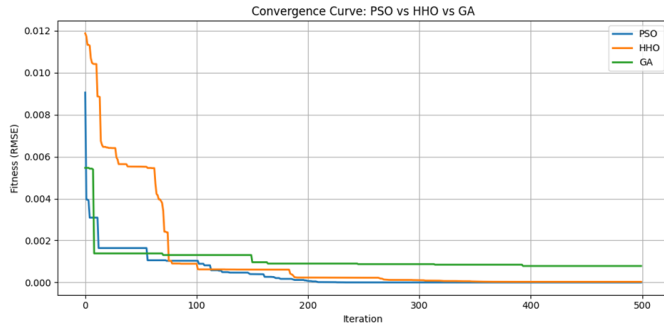


Fig. 3. Convergence curves of PSO, HHO, and GA during optimization.

C. Convergence Behavior

The convergence trends of the three algorithms are shown in Figure 3. PSO reached optimal values in the initial iterations. HHO had a smooth and progressive convergence, ultimately nearing PSO's performance. GA showed slower convergence with higher oscillations but still reached reasonable accuracy.

V. DISCUSSION

This research comparatively evaluated PSO, HHO, and GA, demonstrating the suitability of metaheuristic algorithms for battery parameter estimation, especially when conventional gradient-based methods fail. PSO demonstrated the best convergence speed and accuracy, evidenced by zero error across all metrics. Its adaptive swarm behavior facilitates robust global optimization. HHO exhibited near-zero errors and competitive performance due to its efficient prey-predator-inspired search strategy. It is highly effective for navigating complex nonlinear spaces, making it a strong PSO alternative. GA produced acceptable yet higher errors due to stochastic variation in crossover and mutation. Poor parameter tuning may cause slower convergence. However, GA remains a flexible and reliable option for multi-objective and multimodal problems.

Practical implications:

- PSO: Ideal for real-time and embedded BMS tasks due to fast and accurate convergence.
- HHO: Recommended for offline calibration requiring high accuracy and consistency.

- GA: Suitable for problems involving multiple objectives or diverse solution spaces.

This study supports integrating metaheuristic algorithms into battery modeling workflows. Future work should explore hybrid methods and validate results using real datasets, accounting for temperature and SOC dependencies in BMS design.

VI. CONCLUSION

This study compared the performance of three metaheuristic algorithms—PSO, HHO, and GA—for estimating the parameters R_0 , R_1 , and C_1 of a lithium-ion battery's first-order Thevenin model. Synthetic discharge data was used to evaluate performance through error metrics such as RMSE, MAE, and Max Error. PSO delivered the most accurate results with the least error and best convergence. HHO followed closely with commendable accuracy and faster convergence. GA was successful but exhibited higher prediction error due to reliance on stochastic processes. The convergence curves showed PSO and HHO outperform GA in terms of stability and speed. These findings recommend PSO and HHO for real-time battery modeling. The benchmarking framework introduced here provides researchers and engineers with valuable guidance for selecting optimization techniques in BMS applications. Future studies should validate the approach using experimental data, extend modeling for temperature/aging effects, and explore adaptive and hybrid optimization methods to further enhance performance and reliability.

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